

M. Waseem Ashraf

(385) 286-8600 – waseemashraf1584@gmail.com

EDUCATION

Brigham Young University

Mechanical Engineering PhD.

Jul 2021-Present

Provo, UT

- GPA 3.73 / 4.00
- Research project sponsored by National Science Foundation (**NSF**).
- High throughput atomistic modelling of materials, Kinetics of Materials, Metallurgy, DFT, COMSOL.

Pakistan Institute of Engineering and Applied Sciences

Mechanical Engineering BS

Jun 2014-Jun 2018

Islamabad, PK

- GPA 3.91 / 4.00
- For my final year project, I worked on designing an air ambulance, where I contributed to drag minimization in aerodynamic systems by utilizing Reynolds-Averaged Navier-Stokes (RANS) simulations. My focus was on optimizing flow characteristics to enhance overall performance and efficiency. In addition, I applied topology optimization tools in ANSYS to develop structural designs that maximized material efficiency while improving mechanical integrity, ensuring the system's strength and reliability.
- I participated in the Hyperloop Pod Competition twice, leading Team PIEASloop in designing a pod optimized for minimal drag at ultra-high speeds. My responsibilities included selecting materials and processes, such as determining the shape, grade, and size of neodymium magnets to create a Halbach array for stable eddy current based (passive) levitation at the required speed. This role involved applying principles of materials science, fluid dynamics, and magnetism to improve the system's overall performance and efficiency.
- Learnt material characterization techniques such as Infrared Fourier Transform Spectroscopy (**IFTR**)

EXPERIENCE

Brigham Young University - Graduate Research Assistant

Design & Exploration of Zirconia based Solid-State Actuator

Aug 2021-Present

Provo, UT

- Phase transformation enthalpy calculation based on Quasi-Harmonic Approximation (QHA) using DFPT (VASP) & Phonopy.
- Used SEM to Study Stability of Metal Fluoride Films When Irradiated with Electron Beam. Learnt **Energy Dispersive X-ray Spectroscopy (EDS or EDX)**, Electron Backscatter Diffraction (**EBSD**)
- Orientation Effects on the Electrically Induced Phase Transformation in Zirconia.
- First-principles study of dielectric properties zirconia doped with Ceria, Ytria.
- In addition to characterizing plastic deformation in composite nanomaterial strain sensors, I analyzed the microstructural behavior of the composite material under stress, focusing on failure mechanisms and optimizing the material's properties to improve sensor performance and durability in high precision, in a biomechanic system.
- Teaching assistance for **Engineering Materials: Selection for Design & Science of Engineering Materials**

Lahore University of Management & Sciences

Lab-Instructor

Aug 2018-Aug 2021

Lahore, PK

- Teaching of Experimental Physics lab course.
- Design and development of new experiments.
- Maintenance and up-gradation of equipment and software (MATLAB, LabView).
- Produced manuals for experimental equipment for Physlab.org ([Link1](#), [Link2](#), [Link3](#))

Managerial Tasks

- Simulation trainer for STEM School organized by Innovation Pakistan.

- Appointed Point of Contact (POC) for International Young Physicists' Tournament (IYPT) 2021.

Mari Petroleum Company Limited

Jun 2017-Aug 2017

Engineering Project Intern

Mari, PK

- Supported in the development and engineering evaluation of investment projects, innovation, and improvement of machines in the plant.

Contracted Research and Design Through Fiverr

Aug 2017-Jun 2021

The following are the list of freelance (paid) projects where I performed FEA (Finite Element Analysis) for individuals often hiring out work from their company. Projects ranged in time from three weeks to 4 months.

- **Simulated Human-Centred Designs:** Modelled the specific absorption rate (SAR) for a multi-layered human head under high-frequency electromagnetic waves, optimizing for safety and compliance using COMSOL Multiphysics.
- **Microfluidics Product Design:** Developed and optimized a microfluidic cell counter based on the Coulter Principle, leveraging COMSOL to simulate fluid dynamics and enhance product precision and reliability.
- **Parametric Analysis for Transportation Design:** Optimized passive levitation parameters for advanced transportation systems by simulating geometric and kinematic variations, improving system stability and efficiency through material and process selection aligned with performance requirements.
- **Heat Exchanger Design and Optimization:** In optimizing a shell and tube heat exchanger using ANSYS, I followed **SAE J1995** standards for thermal systems, ensuring the heat exchanger's performance met the required specifications for automotive and aerospace applications, focusing on efficient heat dissipation and material durability.
- **Electromagnetic Simulation:** Explored the behaviour of inductance in internal and external configurations as a function of frequency and distance from coil axis, simulating electromagnetic interactions via COMSOL to guide product development in wireless and sensor technologies.
- **MEMS Product Design:** Designed a micro-electromechanical (MEMS) gripper using COMSOL Multiphysics, focusing on precision and performance for small-scale mechanical applications.
- **Advanced Material Simulation:** Simulated nanoscale mechanical interactions between graphene oxide (GO) hemispheres using elastoplastic finite element methods, providing insights into material behaviour for innovative product applications.
- **Process Optimization for Manufacturing:** Performed process simulation using ANSYS to optimize manufacturing parameters, evaluate material performance, and improve process efficiency, focusing on thermal, structural, and fluid dynamics aspects.
- **Collaborative Medical Device Design:** Partnered with a medical team to model blood flow in a zero-gravity environment, contributing design insights for healthcare devices in extreme conditions.
- **Custom Fluid Handling Device Design:** Designed and built a custom fluid handling device for organ-on-a-chip systems, incorporating CFD simulations and analytical calculations to optimize fluid flow initiation, mimicking arterial pulsatile flow.
- **Diluted Species Flow Simulation:** Modelled the flow of diluted species through serpentine channels using COMSOL, optimizing product design for microfluidic applications.
- **Optical Simulation for Product Testing:** Simulated coupled ray and wave optics to calculate backscattered light intensity in liquid samples with high refractive index spheres, contributing to optical sensor and product testing methodologies.

PUBLICATIONS

- Naheed, M., Ashraf, M. W., Rasheed, M., & Faryad, M. (2021). Computation of specific absorption rate for four-layered human head model exposed to HF electromagnetic waves. In Optical Interactions with Tissue and Cells XXXII (Vol. 11640, pp. 56-60). SPIE.

VOLUNTEER EXPERIENCE

Brigham Young University

Sep 2023-Present

- President Official Mechanical Engineering Graduate Association (OMEGA) 2024-2025.
- Vice-President Official Mechanical Engineering Graduate Association (OMEGA) 2023-2024.

Lahore Science Mela (Science Fair)

Jul 2020-Jul 2021

Full-time volunteer

Lahore, PK

- Designed and built wind tunnel (2m 0.5m 0.5m) to do demonstrations of Bernoulli's principle.

SKILLS

- Languages: English (Fluent), Urdu (Native), Punjabi (Native)
- Software Skills: SolidWorks, PTC CREO, ANSYS, VASP (DFT, DFPT), Atomate, Phonopy
- Programming: C++, MATLAB, Python, Simulink

AWARDS & RECOGNITION

- Recipient of **Ira. A. Fulton Fellowship** (2021-2022)
- One of the finalists for the **Rhodes Scholarships Oxford University** 2018 from Pakistan.
- Awarded certificate of appreciation for best thesis in BS from PIEAS University.
- Awarded merit certificate for academic performance in BS from PIEAS University.
- A lifetime member of Khwarizmi Science Society (KSS).
- Winner glider competition (ASME PIEAS chapter- 2015).